

Lecture 12: Timing Events and Advanced DOM Projects

1. Love Calculator:

A fun calculator that generates a "compatibility score" based on two names. The underlying logic can be anything, this version uses character count.

Logic:

- Take both names as input (e.g., "Akshat" has 6 characters, "Simi" has 4 characters).
- Multiply the character counts, then apply modulus 101 to get a result between 0 and 100.

2. Password Strength Checker:

Validates a password against multiple security conditions in real time.

Logic:

- Attach an input event listener to the password field.
- Read the current value via `.value`.
- Reject immediately if the password is under 8 characters.
- Loop through each character and classify it as uppercase, lowercase, number, or special character.
- The password is "strong" only if all four categories are present.

```
if (password.length < 8) {  
  return;  
}
```

```
let hasCapital = false, hasSmall = false, hasSpecial = false, hasNumber = false;
```

```
for (let i = 0; i < password.length; i++) {  
  let char = password[i];
```

```
  if (char >= 'A' && char <= 'Z')  
    hasCapital = true;
```

```
  else if (char >= 'a' && char <= 'z')  
    hasSmall = true;
```

```
  else if (char >= '0' && char <= '9')  
    hasNumber = true;
```

```
  else hasSpecial = true;  
}
```

```
if (hasCapital && hasSmall && hasNumber && hasSpecial) {  
  // Password passes all criteria  
}
```

Edge Case: Spaces are generally restricted in passwords. External validation libraries are often utilized to handle advanced edge cases and strict pattern matching.

Timing Events in JavaScript:

Property	Description	Use Case
<code>setInterval(callback, ms)</code>	Executes the specified function repeatedly, pausing for the given time interval between each execution.	Live clocks, countdowns.
<code>setTimeout(callback, ms)</code>	Executes the specified function exactly one time after the given delay.	Delayed popups.

3. Live Clock:

Displays the current time, updating every second.

Logic:

- Get the current time and convert it to a readable format using `.toLocaleTimeString()` or `.toLocaleString()`.
- An infinite while loop would freeze the browser, so `setInterval()` is used instead to repeat execution every second.

4. Olympics Countdown Timer:

Calculates and displays the time remaining until a fixed future date.

Logic:

- Get the current timestamp and the target timestamp using `Date.now()` and `.getTime()`.
- Find the difference in milliseconds.
- Convert the remaining milliseconds into days, hours, minutes, and seconds using `Math.floor()` and the modulus operator to carry over the remainder at each step.
- Update the DOM every second using `setInterval()`.

```
let olympicTimestamp = new Date(2028, 6, 14).getTime();
```

```
setInterval(() => {
```

```
  let currentTimeStamp = Date.now();
```

```
  let remainingTime = olympicTimestamp - currentTimeStamp;
```

```
  const days = Math.floor(remainingTime / (1000 * 60 * 60 * 24));
```

```
  remainingTime %= 1000 * 60 * 60 * 24;
```

```
  const hours = Math.floor(remainingTime / (1000 * 60 * 60));
```

```
  remainingTime %= 1000 * 60 * 60;
```

```
  const minutes = Math.floor(remainingTime / (1000 * 60));
```

```
  remainingTime %= 1000 * 60;
```

```
  const seconds = Math.floor(remainingTime / 1000);
```

```
  remainingTime %= 1000;
```

```
  const h2 = document.querySelector('h2');
```

```
  h2.textContent = `${days}:Days ${hours}:Hours ${minutes}:Minutes ${seconds}:Seconds`;  
}, 1000);
```

Practice Projects:

1. Stopwatch:

Build a stopwatch with Start/Resume, Stop, and Reset controls, tracking elapsed time.

2. Astrology / Zodiac Predictor:

Build a predictor that generates personalized outputs based on user input.

- 12 zodiac signs
- A compliment based on the date
- A "victim card" compliment (logic can be anything)
- A recommendation
- A prediction for the future

Note: The above four outputs (compliment, victim card, recommendation, prediction) can be stored as arrays and randomly selected.